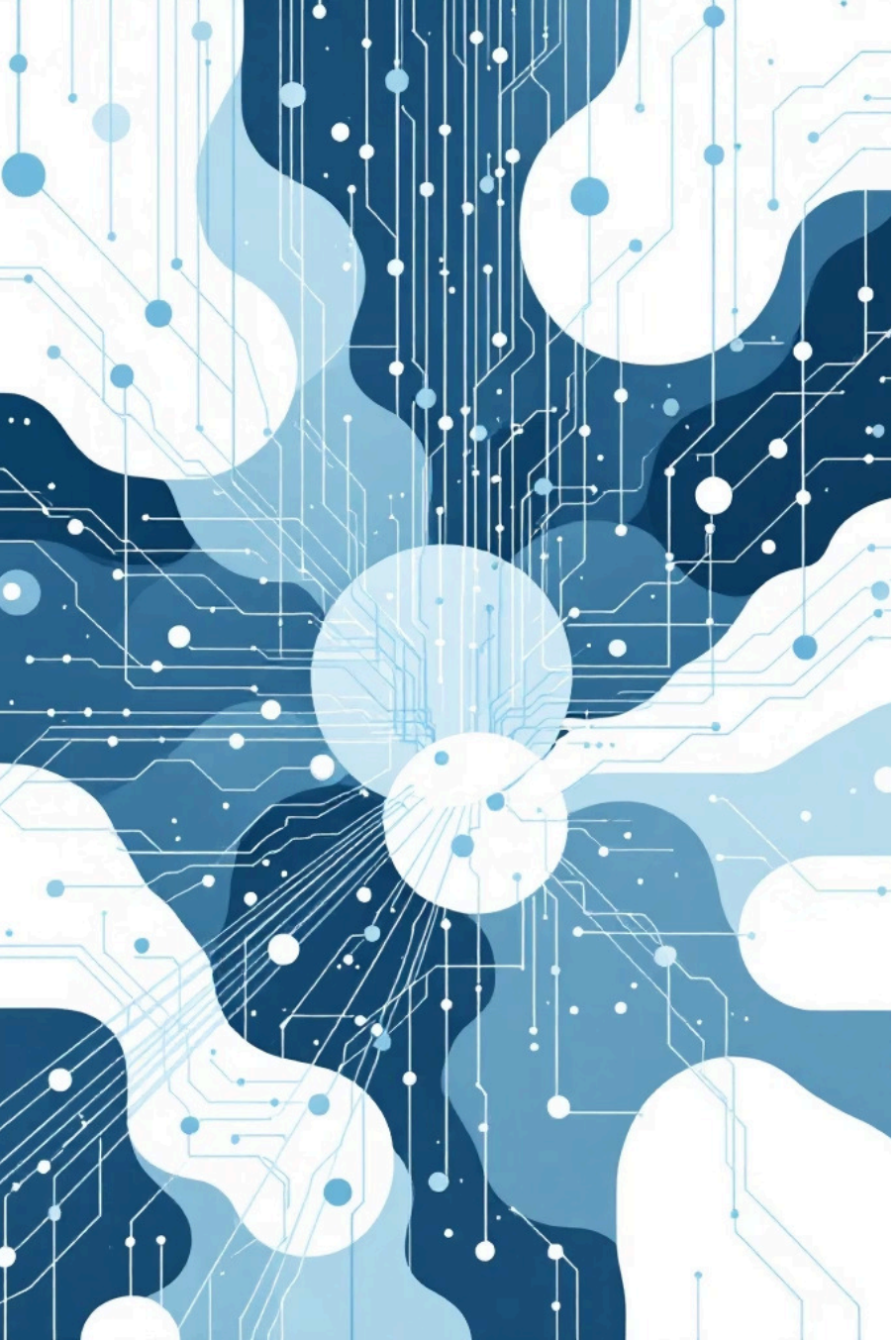


Day 1: Core Concepts of AI & Machine Learning

An exciting introduction to the world of intelligent systems and data-driven learning

Made with GAMMA

Learning Objectives

By the end of Day 1, you will have a solid understanding of the fundamental concepts that power modern AI systems.

Core Concepts

Understand the difference between AI, ML, and Deep Learning

Learning Paradigms

Learn supervised and unsupervised learning approaches

Algorithms

Explore commonly used ML algorithms in industry

Model Building

Build basic machine learning models using Python

Model Evaluation

Interpret model results and evaluation metrics

Made with GAMMA

Today's Journey

Today's session will focus on building a **strong conceptual foundation** of Machine Learning. You'll start with understanding how AI evolved, then learn how machines learn from data, and finally move to implementation.

Our Roadmap

1. The evolution of AI into ML and Deep Learning
2. Understanding how machines learn from data
3. Different types of learning paradigms
4. Major ML algorithms used in industry
5. Simple ML model implementation in Python
6. Evaluating performance using metrics

By the end of today, you'll move from theory to real implementation!

Made with GAMMA

AI, ML & Deep Learning: The Complete Overview

Artificial Intelligence (AI)

AI is the broad domain focused on building intelligent systems that simulate human-like intelligence. It encompasses everything from simple rule-based systems to complex neural networks.

Machine Learning (ML)

ML is a subset of AI where systems learn patterns from data instead of being explicitly programmed with rules. The system improves through experience.

Deep Learning (DL)

DL is a subset of ML that uses multi-layer neural networks for complex pattern recognition. It excels at processing images, sound, and text data.

Made with GAMMA

Supervised Learning

What is Supervised Learning?

Supervised learning is used when your data contains labeled outputs. The model learns from examples where the correct answer is already known.

How It Works

- Training data includes input features and known outputs
- Model learns the mapping between inputs and outputs
- Can predict outcomes for new unseen data
- Requires labeled training data

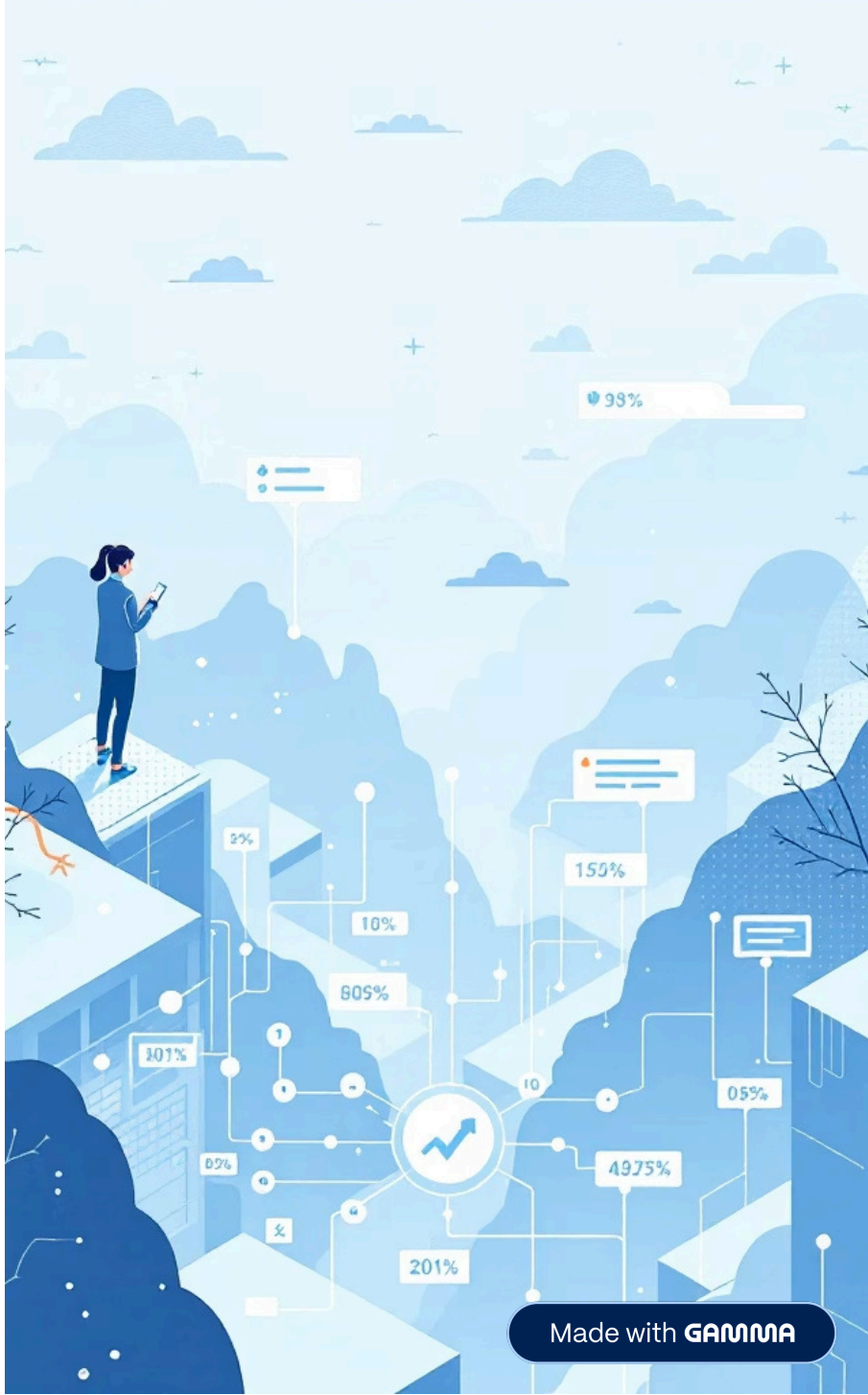
Two Main Types

Regression

Predict continuous values like prices, temperatures, or quantities

Classification

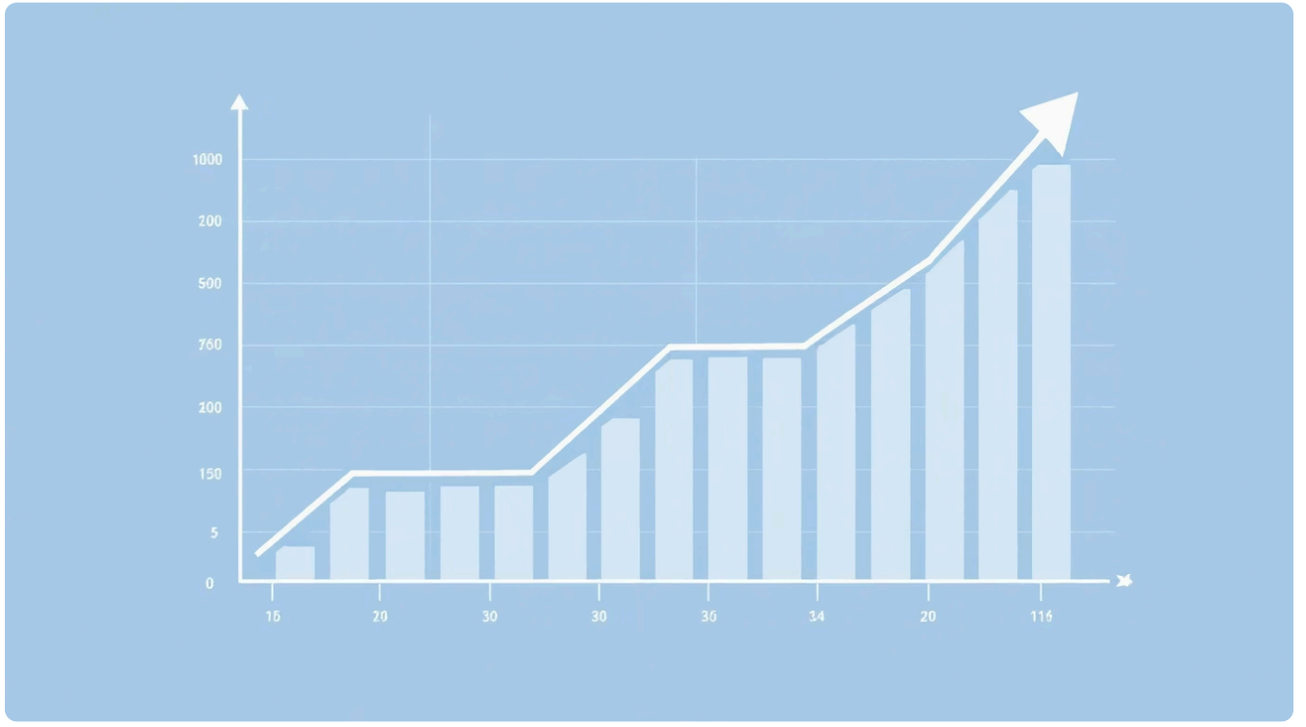
Predict categories like spam/not spam, positive/negative, or image labels



Made with GAMMA

Regression Algorithms

Predicting Continuous Values



Regression algorithms predict continuous values such as prices, temperatures, or quantities. They're essential for forecasting and trend analysis.

- **Linear & Polynomial Regression**

  Models relationships between variables using lines or curves

- **Ensemble Methods**

  Random Forest, Gradient Boosting, XGBoost combine multiple models

- **Regularization**

  Ridge, Lasso, Elastic Net prevent overfitting

- **Other Methods**

  Decision Trees, SVR, KNN for flexible pattern learning

Made with GAMMA

Classification Algorithms

Predicting Categories & Labels

Logistic Regression

Models probability of binary outcomes using S-shaped curves

K-Nearest Neighbors

Classifies based on similarity to neighboring data points

Support Vector Machine

Finds optimal boundary between classes for maximum separation

Naive Bayes

Uses probability theory based on Bayes theorem with independence assumption

Decision Trees & Forests

Creates tree-like rules and ensemble methods for robust predictions

Boosting Algorithms

LightGBM, CatBoost, AdaBoost build models sequentially to correct errors

Made with GAMMA

Unsupervised Learning

Discovering Patterns Without Labels

Unsupervised learning is used when data has no labeled output. The system must find patterns and structure on its own.

Think of it like organizing a messy closet without instructions—you naturally group similar items together (shirts with shirts, pants with pants) without being told what categories to create.

Key characteristics:

- No labeled answers or target outputs provided
- Algorithm discovers hidden patterns and relationships
- Like a detective finding clues without knowing the crime
- Useful for exploration and understanding data structure
- Reveals surprising groupings you might not have expected

Common use cases:

- Customer segmentation (grouping similar shoppers)
- Gene sequencing (finding patterns in DNA)
- Document organization (clustering similar articles)
- Anomaly detection (finding unusual patterns)

Clustering

Groups similar data points together without prior knowledge of groups

Dimensionality Reduction

Compresses data into fewer dimensions while preserving important patterns

Made with GAMMA

Hands-On Activities

Supervised Learning

Implement and evaluate 1 regression model and 1 classification model

Unsupervised Learning

Build and analyze 1 clustering model

Data Preparation

Perform train-test split for reliable model evaluation

Model Evaluation

Calculate and interpret Accuracy and Mean Squared Error metrics

Made with GAMMA